

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

University Theory Examination April 2016

PH405 DOSAGE FORM DESIGN

Date: 14.04.2016, Thursday

Time: 01:30 p.m to 4:30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A What is polymorphism? Explain its significance and methods to identify polymorphism. 10
- B Write a note on suspending agents and emulsifiers. 10
- C Write a note on different types of binders used for pharmaceutical formulations. 10

Q 2 Attempt any TWO of the following

- A (i) Explain importance of particle size and shape in preformulation study. 05
(ii) Explain the principle of wetting in liquid preparation. 05
- B (i) Write a note on coloring agent , sweeteners and flavouring agents. 05
(ii) Write in detail about disintegrants. 05
- C Explain DSC methodology to carry out drug excipients compatibility study with examples. 10

SECTION- II

Q 3 Attempt any TWO of the following

- A Write a detailed note on: ICH guidelines. 10
- B Explain the following pathway of drug degradation with its stabilization techniques. 10
(i) Oxidation (ii) Acid Base catalysis
- C What do you mean by dissolution of drug? Explain the 'Diffusion Layer Model' and 'Surface renewal theory' in detail. 10

Q 4 Attempt any **TWO** of the following

- A** Write a detailed note on: 'Accelerated stability study'. **10**
- B** Explain the physical degradation of drug and write about the preventive measures of it. **10**
- C** Enlist various dissolution test apparatus as per USP. Explain the Type 1 apparatus in detail. **10**
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